

The Mathematics of Money

Chapter 3: Expected Return, Risk, and Diversification

Study Notes Series

1 Expected Return

An investment's future return is uncertain, so it is modeled as a random variable. If an asset can produce return R_i with probability p_i across k possible scenarios, its **expected return** is:

$$E[R] = \sum_{i=1}^k p_i R_i. \quad (1)$$

Expected return tells you the average outcome you would see if the same bet were repeated many times, but it says nothing about how much outcomes vary from one instance to the next. That is the role of risk.

2 Variance and Standard Deviation as Risk

The most common measure of investment risk is the **variance** of returns around the expected value:

$$\sigma^2 = \sum_{i=1}^k p_i (R_i - E[R])^2, \quad (2)$$

and the **standard deviation** $\sigma = \sqrt{\sigma^2}$, which is easier to interpret since it is expressed in the same units as the return itself (e.g., percentage points).

Worked example

Suppose a stock has a 50% chance of returning 20% and a 50% chance of returning -10% in a given year.

$$E[R] = 0.5(0.20) + 0.5(-0.10) = 0.05 = 5\% \quad (3)$$

$$\sigma^2 = 0.5(0.20 - 0.05)^2 + 0.5(-0.10 - 0.05)^2 = 0.0225 \quad (4)$$

$$\sigma = \sqrt{0.0225} = 0.15 = 15\% \quad (5)$$

An expected return of 5% with a 15% standard deviation means actual outcomes will typically swing well above or below that average, reflecting real uncertainty.

3 Diversification: Combining Two Assets

The variance of a portfolio made of two assets with weights w_1 and w_2 (where $w_1 + w_2 = 1$), individual standard deviations σ_1, σ_2 , and correlation coefficient ρ_{12} between their returns, is:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2. \quad (6)$$

The key insight is the cross term. If the two assets are not perfectly correlated ($\rho_{12} < 1$), the portfolio's overall risk is *lower* than the weighted average of the individual risks. When $\rho_{12} = -1$ (perfectly negatively correlated), it is even possible to construct a combination with zero variance, in principle eliminating risk entirely while retaining a positive expected return.

Worked example

Two assets each have $\sigma_1 = \sigma_2 = 20\%$ and are held in equal weights, $w_1 = w_2 = 0.5$.

- If $\rho_{12} = 1$ (perfectly correlated): $\sigma_p = 20\%$, no benefit from diversifying.
- If $\rho_{12} = 0$ (uncorrelated): $\sigma_p^2 = 0.25(0.04) + 0.25(0.04) = 0.02$, so $\sigma_p \approx 14.1\%$, noticeably lower risk for the same expected return.
- If $\rho_{12} = -1$ (perfectly negatively correlated): $\sigma_p = |w_1 \sigma_1 - w_2 \sigma_2| = 0\%$.

This is the mathematical foundation behind the common advice to hold a mix of asset classes (stocks, bonds, real estate, international markets) rather than concentrating in a single investment: as long as their returns are not perfectly correlated, combining them reduces overall portfolio risk without necessarily sacrificing expected return.

Quick check

If two assets are perfectly correlated ($\rho_{12} = 1$), does combining them in a portfolio ever reduce risk below the weighted average of their individual standard deviations? (Answer: no. When $\rho_{12} = 1$, σ_p reduces exactly to $w_1 \sigma_1 + w_2 \sigma_2$, the weighted average, with no diversification benefit.)